

Computer interfacing (CI)

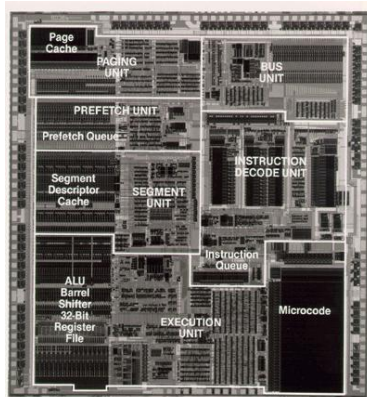
1. Introduction

1.1 Expected knowledge

Among others:

- Use and behavior of electronic components like:
 - *Passive devices*: R, C, L
 - *Semiconductors*: diodes, LED, BJT and MOSFET transistors.
- DC circuit analysis. Ohm's law. Voltages (V), currents (A) and power (W) calculation.
- C Language programming.
- Understanding of technical documentation in english.

1.2 Basic ideas (1): Microprocessor vs microcontroller

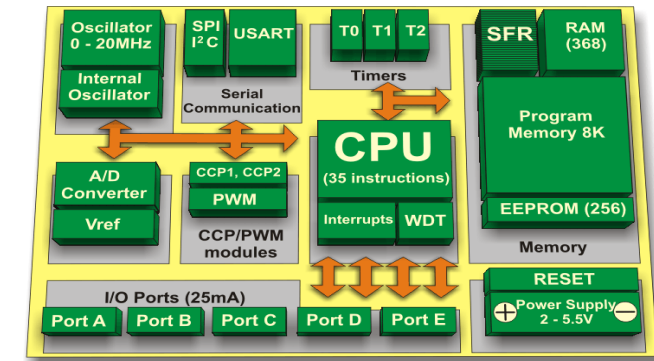


Microprocessor

VS

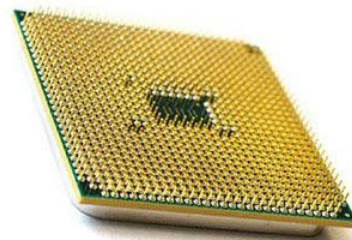


Microcontroller



A **microprocessor** is a CPU in a single chip. Resources like main memory or peripherals are available through an external device (in a PC the mainboard).

A **microcontroller** is a small computer on a single chip, containing a processor core, memory, and programmable input/output peripherals.



Comparison of relative sizes.

1.2 Basic ideas (2): RISC vs CISC architectures.

RISC

Simple instruction set

Regular and fixed instruction format

Wired instructions, take only one or a few clock cycles to execute.

Simple address modes

Separated data and program memory

Most operations are register to register

Take shorter time to design and debug

Provide large number of CPU registers

Simple operations, longer programs
EFFICIENCY?

CISC

Complex instruction set

Irregular instruction format

Complex microcoded instructions, take many clock cycles to execute.

Complex address modes

Combined data and program memory

Most operations can be register to memory

Take longer time to design and debug

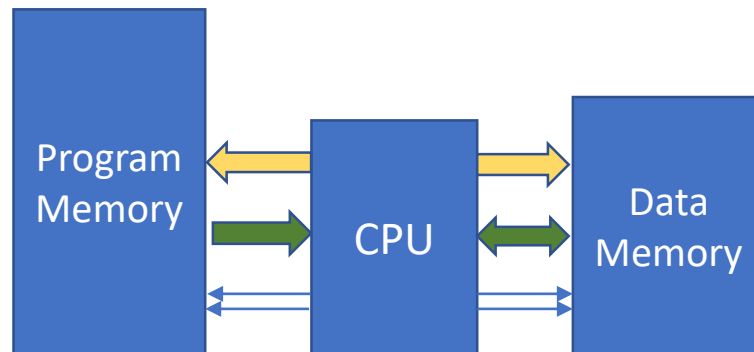
Provide smaller number of CPU registers

Complex operations, shorter programs
EFFICIENCY?

1.2 Basic ideas (3): Harvard vs Von Neumann Archs.

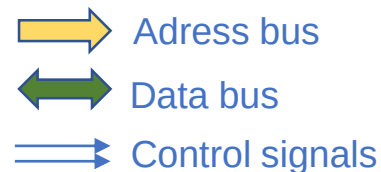
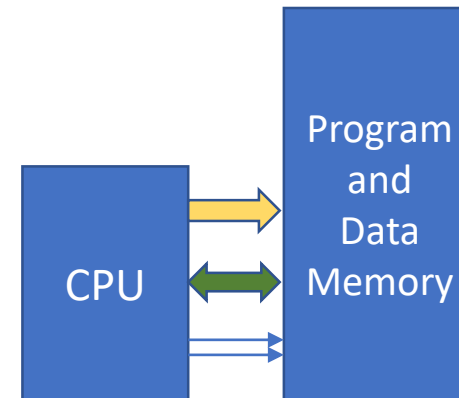
Harvard Architecture

Is a computer architecture with separate storage and signal pathways for instructions and data.



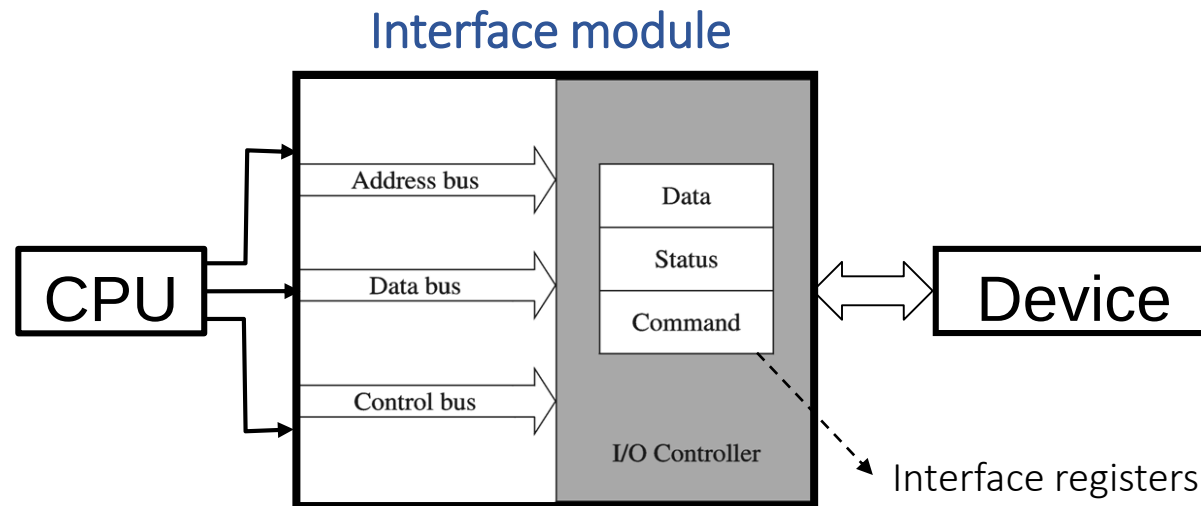
Von Neumann Architecture

Instructions and data are stored in the same memory, so instructions are fetched over the same data path used to fetch data.



1.2 Basic ideas (4): I/O Registers and Memory Space.

All devices (keyboards, audio, screens...) are managed through **interfaces**, when there is an Operating System, drivers hide this interfaces to the users.



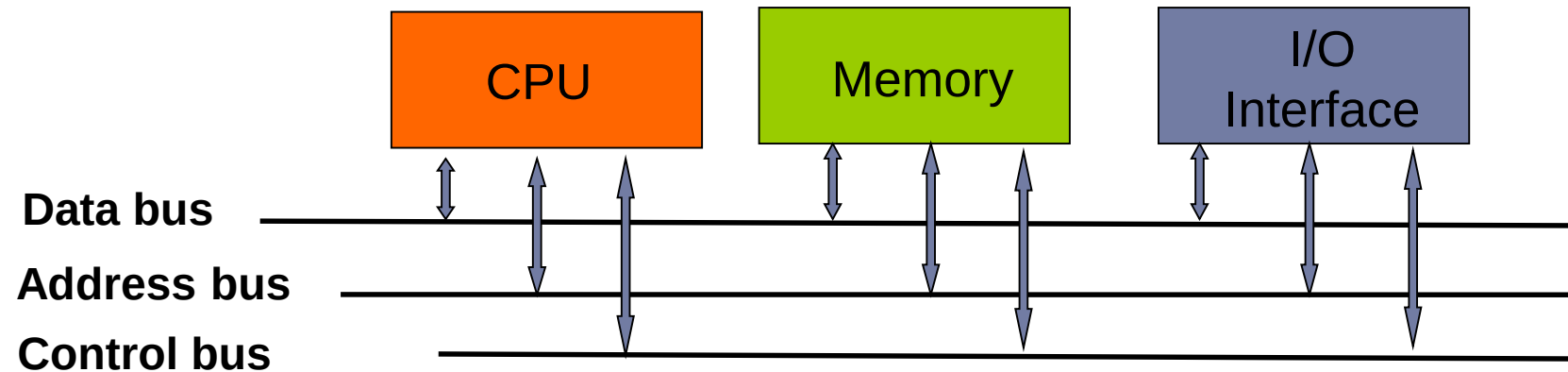
Digital interface registers are also called **PORTS**

System/microcontroller interfaces are programmed with **registers** (command, data...)

Where are Data, Status and Command registers located?

1.2 Basic ideas (4): I/O Registers and Memory Space.

If the same address bus is used for both memory and I/O, **how should the hardware be designed to differentiate between memory and I/O reads and writes?**



1.2 Basic ideas (4): I/O Registers and Memory Space.

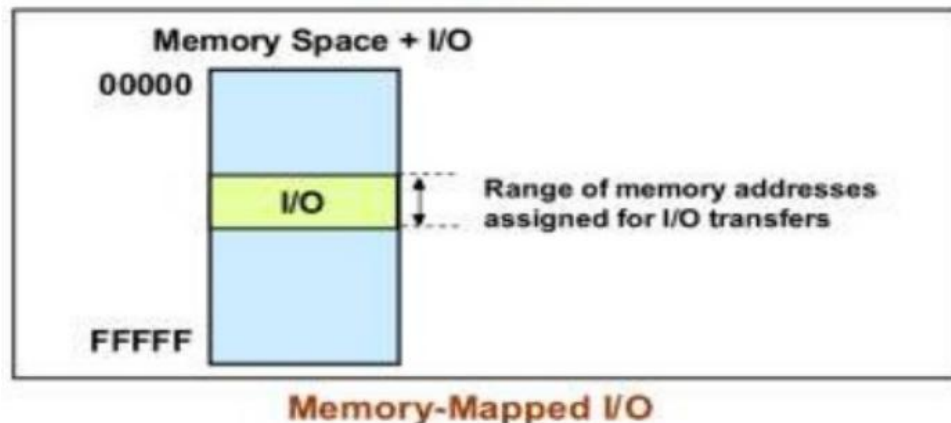
Solution 1: Memory Mapped I/O (MOTOROLA)

Any instruction that reads or writes memory can read/write I/O Port.

Address specifies which module (input, output, RAM, ROM), will communicate with the processor.

Example:

```
LDAA $200A    ;load value from address $200A
STAA $0024    ;store value to Port address $0024
```



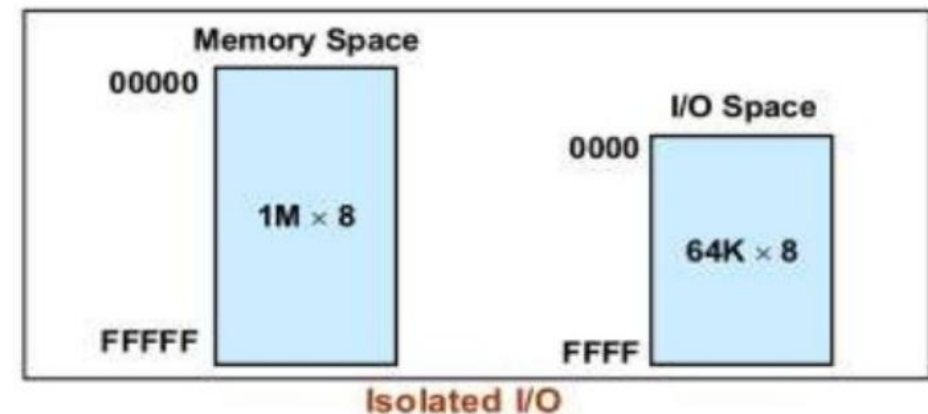
Solution 2: Isolated I/O (INTEL)

The control bus signals that activate the I/O are separate from those that activate the memory device.

These systems have a separate address space. Separate instructions are used to access I/O and Memory.

Example:

```
IN AL,$10     ;read from port $10
MOV $200A,AL  ;store value to RAM address $200A
```



1.2 Basic ideas (5): Memory technologies

Main classification

Volatile: loses data when power is turned off.

Typical volatile: Random-Access Memory (**RAM**). It's a Read/Write memory.

Non-volatile: keeps stored data when power is turned off.

Typical non-volatile: Read-Only Memory (**ROM**). Can only be read but not written by the processor.

1.2 Basic ideas (5): Memory technologies

Types of RAM:

Dynamic random-access memory (DRAM): need periodic refresh.

- Capacitor technology.
- Big capacities (8GB, 32GB).
- Use as PC's main memory

Static random-access memory (SRAM): no periodic refresh is required

- More power consuming. Transistor technology.
- Fast and simple.
- Smaller capacities (KB—MB)
- Use as microcontroller's memory, PC registers, PC Caches.

1.2 Basic ideas (5): Memory technologies

Types of ROM:

Mask-programmed read-only memory (MROM): programmed when being manufactured.

Programmable read-only memory (PROM): can be programmed by the end user (OTP).

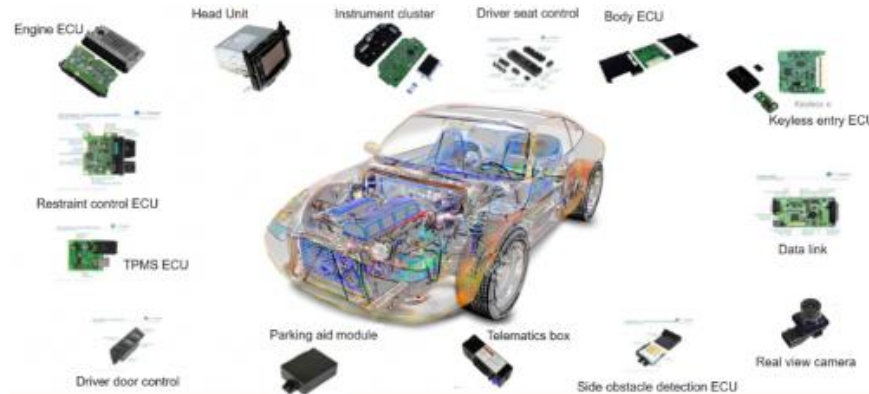
Electrically erasable programmable ROM (EEPROM)

- electrically programmable many times
- electrically erasable many times
- can be erased one location, one row, or whole chip in one operation

Flash memory (e.g. pendrive, SSD disks)

- electrically programmable many times
- electrically erasable many times
- can only be erased in bulk (either a block or the whole chip)

1.3 Use of microcontrollers



Car microcontrollers



Drone microcontrollers



IoT (Internet of things)

The global microcontroller market size was estimated at USD 20.61 billion in 2022 and is expected to reach USD 22.73 billion in 2023.

1.3 Use of microcontrollers: Setup for CI



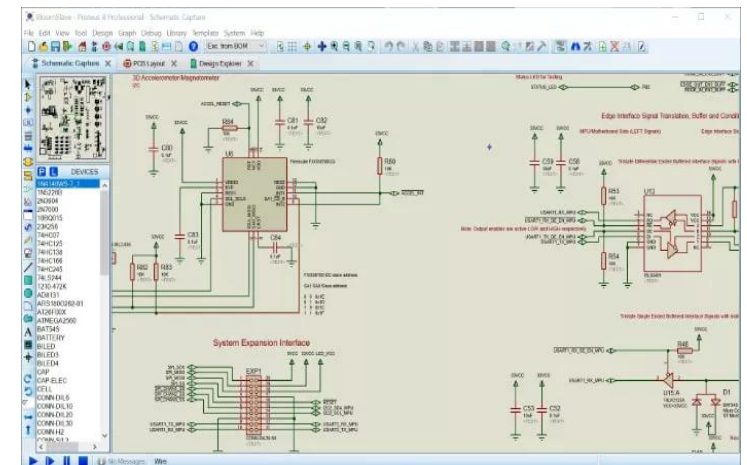
Sellings of 8,3 billion dollars in 2023.
22500 employees.
Leader in the microcontroller industrial sector.



PIC 18F45K22,
microcontroller device



Mikroe: EasyPIC 7,
development board.



Proteus: Hardware &
Code simulation.

1.3 Use of microcontrollers: Setup for CI



PIC 18F45K22,
microcontroller device

RISC architecture.
Harvard Architecture.
Memory mapped I/O registers.
FLASH memory to store programs.
EEPROM memory for user data.
SRAM memory for registers and data RAM.

Entregable N° 1

Q1 Un microcontrolador amb l'entrada/sortida mapejada en la memòria RAM serà més aviat RISC o CISC?

Q2 Un xip microcontrolador que pretén ser de baix consum tindrà molta, poca RAM o no té res a veure?

Q3 Si en la ROM es guarda el programa, en la RAM es guarden els valors de les variables, llavors on es guarden els valors de les comandes, consignes i paràmetres?